Recitation 2: Proof Techniques

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Warmup

1. For every real number x, $x^2 \geq x$
2. Prove that if n is an odd integer, then $3n + 7$ is an even integer.
3. Prove that there is no smallest positive rational number.

Proof by Contraposition

1. Let $n \in \mathbb{Z}$. If $(3n + 2)^2$ is even, then n is even.
2. Prove that if ab is even (with $a, b \in \mathbb{Z}$), then a is even or b is even.
3. If $n = ab$, where $a, b \in \mathbb{Z}$, then either $a \leq \sqrt{n+1}$ or $b \leq \sqrt{n+1}$.

Proof by Contradiction

1. Prove that if $r \in \mathbb{Q}$ and $s \notin \mathbb{Q}$, then $r + s \notin \mathbb{Q}$.
2. Prove that there are no integers x and y such that $x^2 = 8y + 2$.

Proof by Induction

1. Prove that for all $n \geq 1$, $n! \geq 2^{n-1}$.
2. Prove that for all $n \geq 1$, $10^n - 1$ is divisible by 9, i.e., show that $9 \mid (10^n - 1)$.

Proof by Parts

1. Prove that there is no integer x such that $x^2 \equiv 2 \pmod 3$.
2. Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is odd, then $a + b$ is also odd.